

# The Complete Guide to Shavian, Tengwar, and Featural Transliteration

## Part 1: The Shavian Alphabet & ReadLex Specification

The Shavian alphabet is a purely phonemic writing system designed by Kingsley Read to fulfill the bequest of George Bernard Shaw. It replaces traditional non-phonetic English spelling with a 1:1 sound-to-letter mapping.

### 1. Character Taxonomy

Shavian characters fall into four structural categories based on vertical height and composition:

- **Tall (Voiceless Consonants):** Extend above the x-height.
- **Deep (Voiced Consonants):** Extend below the baseline.
- **Short (Vowels, Liquids, Nasals):** Fit entirely within the x-height.
- **Compound (Vocalic Ligatures):** Combine rhotic (/r/) sounds and vowel clusters into single glyphs.

### Consonant Sound Pairs

Tall and Deep consonants form physical pairs where rotating or flipping a Tall (voiceless) letter generates its Deep (voiced) counterpart.

| Sound Pair | Tall Glyph<br>(Voiceless) | Deep Glyph<br>(Voiced) | Phonetic<br>Names |
|------------|---------------------------|------------------------|-------------------|
| /p/ – /b/  | ᵛ                         | ᵇ                      | peep – bib        |
| /t/ – /d/  | ᵀ                         | ᵀ̣                     | tot – dead        |
| /k/ – /g/  | ᵀ̣                        | ᵇ                      | kick – gag        |
| /f/ – /v/  | J                         | ᵀ̣                     | fee – vow         |
| /θ/ – /ð/  | ᵀ̣                        | ᵀ̣̣                    | thigh – they      |
| /s/ – /z/  | S                         | ᵀ̣̣                    | so – zoo          |

| Sound Pair     | Tall Glyph<br>(Voiceless) | Deep Glyph<br>(Voiced) | Phonetic<br>Names |
|----------------|---------------------------|------------------------|-------------------|
| /ʃ/ – /ʒ/      | ℓ                         | ɹ                      | sure –<br>measure |
| /tʃ/ –<br>/dʒ/ | ℓ                         | ɹ                      | church –<br>judge |
| /h/ – /w/      | ɸ                         | /                      | he – woe          |

## Vowels, Liquids, and Compound Ligatures

| Category                    | Characters & Keywords                                                                                                 |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Liquids &amp; Nasals</b> | ℓ (/l/ loll), ʃ (/m/ mime), ʌ (/n/ nun), ɹ (/ŋ/ hung), ɔ (/r/ roar), ʌ (/j/ yea)                                      |
| <b>Short Vowels</b>         | ɪ (/ɪ/ if), ʌ (/ɛ/ egg), ʌ (/æ/ ash), ɹ (/ə/ ado), ɹ (/ʌ/ up), ʌ (/ɒ/ on), ʌ (/ʊ/ wool)                               |
| <b>Long Vowels</b>          | ɪ (/i:/ eat), ɹ (/eɪ/ age), ɹ (/ɑ:/ ah), ɹ (/ɔ:/ awe), ʌ (/oʊ/ oak), ʌ (/u:/ ooze)                                    |
| <b>Diphthongs</b>           | ɹ (/aɪ/ ice), ɹ (/ɔɪ/ oil), ɹ (/aʊ/ out)                                                                              |
| <b>Compound Ligatures</b>   | ɹ (/ɑ:r/ are), ɹ (/ɔ:r/ or), ɹ (/ɛər/ air), ɹ (/ɜ:r/ err), ɹ (/ər/ array), ɹ (/ɪər/ ear), ɹ (/iə/ Ian), ʌ (/ju:/ yew) |

## 2. The 12 ReadLex Orthographic Rules

- Strict Sound-to-Letter Phonetics:** Write strictly by spoken sound, ignoring silent Latin letters (e.g., *write* = ɹɹɪ).
- Phonetic Voicing Assimilation in Suffixes:** Suffixes match spoken voicing (e.g., *cats* = ʌɹɪʃ vs. *dogs* = ɹɹɹ; *walked* = ɹɹʌɪ vs. *played* = ɹɹɹɹ).
- The Big Five Abbreviations:** Five high-frequency function words are abbreviated as single shorthand letters: *the* = ʌ, *of* = ʃ, *and* = ʌ, *to* = 1, *for* = J.
- Monosyllabic Invariance:** Single-syllable words maintain their fully stressed spelling regardless of context (e.g., *but* is always ɹɹɪ).
- Monosyllabic Weak Vowel Exception:** Indefinite articles are explicitly written weak using schwa (ɹ): *a* = ɹ, *an* = ɹʌ.

6. **Stress Distinction in Polysyllabic Words:** Vowel reduction reflects stress shifts (e.g., noun *convict* = ɔ̃ʌʋɪɔ̃ vs. verb *convict* = ɔ̃ʌʋɪɔ̃).
  7. **No Letter Gemination:** Consonants are never doubled unless articulated twice across syllable boundaries (e.g., *annoy* = ʌʌ vs. *unnamed* = ʌʌʌʌ).
  8. **Syllabic Consonant Formatting:** Insert a schwa (ɪ) before terminal liquids or nasals (e.g., *little* = ɔ̃ʌʌɪ, *driven* = ɔ̃ʌʌʌ).
  9. **Verbal Contraction Formatting:** Negative contractions insert ɪ before ʌ and omit apostrophes (e.g., *didn't* = ɔ̃ʌʌʌ).
  10. **Proper Noun Marking:** Capital letters are omitted entirely; proper nouns use a middle Namer Dot (·) immediately preceding the word (e.g., *London* = ·ɔ̃ʌʌʌ).
  11. **Unstressed Terminal Vowels (happy Rule):** Words ending in unstressed -y or -ie use ɪ (e.g., *happy* = ʌʌʌ). Tense, fully stressed final vowels use ɪ (e.g., *trustee* = ʌʌʌʌ).
  12. **Cross-Dialectal "Rhotic RP" Standard:** Standard spelling uses a broad rhotic Received Pronunciation base so rhotic and non-rhotic speakers share a single format.
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## Part 2: Tolkien's Tengwar & CSUR Standard

Tengwar is a featural abugida created by J.R.R. Tolkien. Primary consonants (*tengwar*) represent structural articulation features, while vowels (*tehtar*) sit above or below consonant bases as diacritics.

### 1. The Fëanorian Consonant Grid (Appendix E Layout)

In Tolkien's standard layout, primary consonants are organized into **4 Series** (Columns: Place of Articulation) and **6 Grades** (Rows: Manner of Articulation and Voicing):

- **Series I (Témar 1):** Dental / Alveolar (/t/, /d/, /θ/, /ð/)
- **Series II (Témar 2):** Labial / Bilabial (/p/, /b/, /f/, /v/)
- **Series III (Témar 3):** Palatal / Affricate (/tʃ/, /dʒ/, /ʃ/, /ʒ/)
- **Series IV (Témar 4):** Velar (/k/, /g/, /ŋ/)

| Grade / Manner                                                          | Series I<br>(Dental)                         | Series II<br>(Labial)                     | Series III<br>(Palatal)                | Series IV<br>(Velar)                       |
|-------------------------------------------------------------------------|----------------------------------------------|-------------------------------------------|----------------------------------------|--------------------------------------------|
| <b>Grade 1: Voiceless Stops</b> ( <i>Stem Down, Single Loop</i> )       | p<br>U+E000<br><br><i>Tinco</i><br>(/t/)     | p<br>U+E001<br><br><i>Parma</i><br>(/p/)  | q<br>U+E002<br><br><i>Calma</i> (/tʃ/) | q<br>U+E003<br><br><i>Quesse</i><br>(/k/)  |
| <b>Grade 2: Voiced Stops</b> ( <i>Stem Down, Double Loop</i> )          | ɸ<br>U+E004<br><br><i>Ando</i> (/d/)         | ɸ<br>U+E005<br><br><i>Umbar</i><br>(/b/)  | ɸ<br>U+E006<br><br><i>Anga</i> (/dʒ/)  | ɸ<br>U+E007<br><br><i>Ungwe</i><br>(/g/)   |
| <b>Grade 3: Voiceless Fricatives</b> ( <i>Stem Up, Single Loop</i> )    | h<br>U+E008<br><br><i>Thúle</i> (/θ/)        | h<br>U+E009<br><br><i>Formen</i><br>(/f/) | d<br>U+E00A<br><br><i>Harma</i> (/ʃ/)  | d<br>U+E00B<br><br><i>Hwesta</i><br>(/hw/) |
| <b>Grade 4: Voiced Fricatives</b> ( <i>Stem Up, Double Loop</i> )       | ħ<br>U+E00C<br><br><i>Anta</i> (/ð/)         | ħ<br>U+E00D<br><br><i>Ampa</i> (/v/)      | cd<br>U+E00E<br><br><i>Anca</i> (/ʒ/)  | cd<br>U+E00F<br><br><i>Unque</i><br>(/gh/) |
| <b>Grade 5: Nasals</b> ( <i>Short Stem, Double Loop</i> )               | ɹ<br>U+E010<br><br><i>Númen</i><br>(/n/)     | ɹ<br>U+E011<br><br><i>Malta</i><br>(/m/)  | ɸ<br>U+E012<br><br><i>Noldo</i> (/ŋ/)  | ɸ<br>U+E013<br><br><i>Nwalme</i><br>(/nw/) |
| <b>Grade 6: Glides / Semi-vowels</b> ( <i>Short Stem, Single Loop</i> ) | ɹ<br>U+E014<br><br><i>Óre</i> (/r/<br>final) | ɹ<br>U+E015<br><br><i>Vala</i> (/w/)      | ɸ<br>U+E016<br><br><i>Anna</i> (/j/)   | ɸ<br>U+E017<br><br><i>Vilya</i>            |

## 2. Additional & Irregular Tengwar

Beyond the standard 24-tengwa grid, specialized base characters handle liquids, sibilants, glottal sounds, and silent carriers:

| Tengwa Glyph | CSUR Hex | Tengwa Name           | Structural Description / Function                      |
|--------------|----------|-----------------------|--------------------------------------------------------|
| ṛ            | U+E018   | Rómen                 | Trilled/vocalic /r/ (used before vowels)               |
| ḷ            | U+E01A   | Lamba                 | Upward hooked liquid character (/l/)                   |
| ṣ            | U+E01C   | Silme                 | Upright S-curve sibilant (/s/)                         |
| ḥ            | U+E01D   | Silme<br>Nuquerna     | Inverted S-curve (used when carrying top tehta)        |
| ḏ            | U+E01E   | Esse / Aze            | Z-curve / inverted double loop (/z/)                   |
| Ḑ            | U+E01F   | Esse Nuquerna         | Inverted Z-curve (used when carrying top tehta)        |
| ḡ            | U+E020   | Hyarmen               | Downward stem with left hook (/h/)                     |
| λ            | U+E028   | Telco (Short Carrier) | Unadorned vertical pillar carrying short orphan vowels |
| ḏ            | U+E029   | Ára (Long Carrier)    | Extended vertical pillar carrying long orphan vowels   |

### 3. Vowel Tehtar & Diacritic Marks

In English General Mode, vowel diacritics (*tehtar*) sit directly above or below the following consonant base glyph:

| Tehta Mark | CSUR Hex | Tehta Name  | Phonetic Mapping | Visual Placement                    |
|------------|----------|-------------|------------------|-------------------------------------|
| ˆ          | U+E040   | a-tehta     | /æ/, /ɑː/, /aɪ/  | Three dots in a triangle above base |
| ˙          | U+E044   | i-tehta     | /ɪ/, /iː/        | Single dot above base               |
| ˘          | U+E045   | schwa-tehta | /ə/ (schwa)      | Single dot <b>below</b> base        |

| Tehta Mark | CSUR Hex | Tehta Name          | Phonetic Mapping     | Visual Placement                   |
|------------|----------|---------------------|----------------------|------------------------------------|
| ˊ          | U+E046   | <i>e-tehta</i>      | /ɛ/, /eɪ/            | Single acute stroke above base     |
| ˊˊ         | U+E047   | <i>long e-tehta</i> | /iː/ (FLEECE)        | Double acute stroke above base     |
| ˋ          | U+E04A   | <i>o-tehta</i>      | /ɒ/, /ɔː/, /oʊ/      | Right-facing curl above base       |
| ˋˋ         | U+E04C   | <i>u-tehta</i>      | /ʌ/, /ʊ/, /uː/       | Left-facing curl above base        |
| ˋˋˋ        | U+E04D   | <i>long u-tehta</i> | /uː/ (GOOSE)         | Double left-facing curl above base |
| ˘          | U+E04E   | <i>nasal bar</i>    | Preconsonantal nasal | Horizontal tilde above base stop   |

## Part 3: The Shavian-to-Tengwar Transliteration Bridge

By using Shavian as an Intermediate Representation (IR), transliteration bypasses non-phonetic English orthography entirely, converting text through a 5-pass state machine.

### 1. Direct Phoneme Mapping Matrix

| Shavian Glyph | IPA Phoneme | Target Tengwa / Tehta | CSUR Hex Stream | State Machine Transformation Rule |
|---------------|-------------|-----------------------|-----------------|-----------------------------------|
| ᑭ             | /p/         | <i>Parma</i>          | U+E001          | Direct 1:1 Consonant Base         |
| ᑭ̌            | /b/         | <i>Umbar</i>          | U+E005          | Direct 1:1 Consonant Base         |
| ᑭ̎            | /t/         | <i>Tinco</i>          | U+E000          | Direct 1:1 Consonant Base         |
| ᑭ̏            | /d/         | <i>Ando</i>           | U+E004          | Direct 1:1 Consonant Base         |
| ᑭ̐            | /k/         | <i>Quesse</i>         | U+E003          | Direct 1:1 Consonant Base         |

| Shavian Glyph | IPA Phoneme | Target Tengwa / Tehta   | CSUR Hex Stream | State Machine Transformation Rule                              |
|---------------|-------------|-------------------------|-----------------|----------------------------------------------------------------|
| ᑭ             | /g/         | <i>Ungwe</i>            | U+E007          | Direct 1:1 Consonant Base                                      |
| J             | /f/         | <i>Formen</i>           | U+E009          | Direct 1:1 Consonant Base                                      |
| ƒ             | /v/         | <i>Ampa</i>             | U+E00D          | Direct 1:1 Consonant Base                                      |
| ᑭ             | /θ/         | <i>Thúle</i>            | U+E008          | Direct 1:1 Consonant Base                                      |
| ᑭ             | /ð/         | <i>Anta</i>             | U+E00C          | Direct 1:1 Consonant Base                                      |
| S             | /s/         | <i>Silme / Nuquerna</i> | U+E01C / U+E01D | Flips to <i>Silme Nuquerna</i> ( U+E01D ) if top tehta present |
| z             | /z/         | <i>Esse / Nuquerna</i>  | U+E01E / U+E01F | Flips to <i>Esse Nuquerna</i> ( U+E01F ) if top tehta present  |
| ı             | /ɪ/         | <i>i-tehta</i>          | U+E044          | Attach dot above next available consonant                      |
| ɛ             | /ɛ/         | <i>e-tehta</i>          | U+E046          | Attach acute stroke above next available consonant             |
| ʌ             | /æ/         | <i>a-tehta</i>          | U+E040          | Attach three dots above next available consonant               |
| ɐ             | /ə/         | <i>schwa-tehta</i>      | U+E045          | Attach single dot <b>below</b> next available consonant        |
| ʊ             | /ɜɪr/       | <i>Rómen + u-tehta</i>  | U+E018 U+E04C   | NURSE vowel: Decomposes into <i>Rómen + u-tehta</i>            |
| ʊ             | /ɛər/       | <i>Rómen + e-tehta</i>  | U+E018 U+E046   | SQUARE vowel: Decomposes into <i>Rómen + e-tehta</i>           |
| ʊ             | /ər/        | <i>Óre</i>              | U+E014          | letter vowel: Decomposes into unadorned <i>Óre</i>             |
| ʊ             | /aɪ/        | <i>Anna + a-tehta</i>   | U+E016 U+E040   | PRICE diphthong: Offglide <i>Anna</i> base + <i>a-tehta</i>    |

| Shavian<br>Glyph | IPA<br>Phoneme | Target<br>Tengwa /<br>Tehta | CSUR Hex<br>Stream | State Machine<br>Transformation Rule                |
|------------------|----------------|-----------------------------|--------------------|-----------------------------------------------------|
| o                | /oʊ/           | Vala + o-<br>tehta          | U+E015<br>U+E04A   | GOAT diphthong:<br>Offglide Vala base + o-<br>tehta |

## 2. Verified Test Cases & Output Comparison

To verify visual output in Obsidian or PDF export, install [Alcarin Tengwar](#). The table below lists the exact Shavian input, rendered Tengwar output, CSUR hex bytes, and feature rules applied:

| Word    | Shavian<br>Input | CSUR<br>Tengwar<br>Output | CSUR Hex<br>Stream                              | Applied Rule / Logic                                                                                   |
|---------|------------------|---------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| winter  | /ʍɪnɪr/          | 𐌵𐌶𐌵𐌶𐌵                     | U+E015 U+E010<br>U+E044 U+E000<br>U+E014        | <i>i-tehta</i> attaches to<br><i>Númen</i> ; <i>Tinco</i><br>follows; ends in <i>Óre</i> .             |
| chamber | ʧɛmɪbɛr/         | 𐌵𐌶𐌵𐌶𐌵                     | U+E002 U+E016<br>U+E046 U+E005<br>U+E04E U+E014 | Nasal pair 𐌶𐌵 lacks<br>vowel, emitting<br><i>Umbar</i> + <i>Nasal Bar</i><br>Above ( U+E04E ).         |
| third   | ðɜrð/            | 𐌵𐌶𐌵𐌶𐌵                     | U+E008 U+E018<br>U+E04C U+E004                  | NURSE vowel 𐌶<br>decomposes into<br><i>Rómen</i> + <i>u-tehta</i><br>( U+E04C ).                       |
| hair    | hɛr/             | 𐌵𐌶𐌵                       | U+E020 U+E018<br>U+E046                         | SQUARE vowel 𐌶<br>decomposes into<br><i>Rómen</i> + <i>e-tehta</i><br>( U+E046 ).                      |
| bruised | bruɪzd/          | 𐌵𐌶𐌵𐌶𐌵                     | U+E005 U+E018<br>U+E01F U+E04D<br>U+E004        | Carrying double u-<br>curl forces <i>Esse</i> to<br>flip to <i>Esse</i><br><i>Nuquerna</i> ( U+E01F ). |
| merry   | mɛrɪ/            | 𐌵𐌶𐌵𐌶𐌵                     | U+E011 U+E018<br>U+E046 U+E028<br>U+E044        | Word-final short<br>vowel attaches to<br><i>Short Carrier</i><br>( <i>Telco</i> , U+E028 ).            |

| Word          | Shavian Input | CSUR Tengwar Output | CSUR Hex Stream                                                  | Applied Rule / Logic                                                    |
|---------------|---------------|---------------------|------------------------------------------------------------------|-------------------------------------------------------------------------|
| <i>he</i>     | ȝȥ            | ȝđ                  | U+E020 U+E029<br>U+E044                                          | Word-final long vowel attaches to <i>Long Carrier</i> (Ára, U+E029).    |
| <i>lady's</i> | ccȥȥ'ȝ        | đáȥȥ'ccȥ            | U+E01A U+E016<br>U+E046 U+E004<br>U+E028 U+E044<br>U+2019 U+E01E | Contraction apostrophe splits token; trailing <i>Esse</i> is unadorned. |